

Research Idea: PJ0209

ALMA Thesis Planner (automated pipeline)

Resolved CO line-ratio mapping of one Planck-selected lensed starburst: a magnification-robust, region-based non-LTE excitation analysis

Motivation. The parent program (ADS/JAO.ALMA#2024.1.01396.S) targets 8 Planck-selected lensed starbursts at $z = 1\text{--}3$ with apparent $\log(L_{\text{IR}}) = 13\text{--}14$ L_{sun} , spanning four orders of magnitude in relative high-J vs. mid-J line ratios, with 3–4 CO transitions each drawn from $J = 2\text{--}15$ (plus [CI]) at $0.1\text{--}0.2''$ resolution and $>10\text{--}20$ beams across the emission. The program’s stated end goal — simultaneous pixel-by-pixel, turbulence-driven non-LTE radiative transfer of line and continuum in the reconstructed source plane, for all 8 sources — is a multi-year, full-paper effort. This thesis takes the first, self-contained scientific step: does the extreme *global* high-J/mid-J ratio of a single source reflect a spatially uniform excitation state, or is it produced by a compact, highly excited component embedded in a more extended, lower-excitation reservoir?

Core idea and why it is tractable. For emission that is co-spatial on the sky, the lensing magnification is a common multiplicative factor at a given image-plane position and therefore *cancels in a line ratio* measured at that position. This means resolved *ratio* maps carry intrinsic excitation information without requiring a lens model or source-plane inversion — the single most expensive piece of the parent analysis is deliberately removed from scope. Differential magnification between components of different intrinsic size does not cancel, and quantifying that caveat is itself part of the thesis rather than an afterthought.

Scope (one student, one to two semesters). 1. **Target selection (weeks 1–3).** From the 8 program sources, select one (with a second held as backup) on the basis of delivered data quality: highest continuum/line S/N, most complete J-coverage, and the most extended or multiply-imaged morphology. All subsequent work is on this one source. 2. **Matched imaging (weeks 3–10) — the technical heart.** Re-image the 3–4 observed transitions from the calibrated measurement sets in CASA, enforcing (a) a common uv -range across transitions so each cube is sensitive to the same range of angular scales, and (b) a common circular restoring beam via tapering/smoothing. Because the transitions span a wide range in J and hence sky frequency, the native beams and uv coverage differ substantially; skipping this step is the dominant systematic in resolved ratio work. Continuum is estimated from line-free channels and subtracted, with the continuum image itself retained as a product. 3. **Ratio maps (weeks 10–16).** Produce matched-velocity moment-0 maps and pixel-wise ratio maps (high-J/mid-J, and CO/[CI] where both are available), with full noise propagation and explicit S/N masking. Test measurement robustness by comparing ratios between multiply-imaged/arc positions that map to the same intrinsic region, which should agree within errors if the reduction is unbiased. 4. **Region-based non-LTE modelling (weeks 16–26).** Rather than fitting every pixel, extract $\sim 5\text{--}10$ physically motivated apertures (peak, arc midpoints, outskirts, and a global aperture for comparison with the already-constrained global gas properties) and fit each with a coarse grid of one-zone escape-probability / large-velocity-gradient non-LTE models to constrain kinetic temperature, H₂ volume density, and CO column per unit velocity

gradient. Degeneracies are mapped explicitly (likelihood surfaces, not single best-fit points); with only 3–4 transitions per source the honest deliverable is a constrained *region* of parameter space per aperture, and demonstrating where the data do and do not break the $T_{\text{kin}}\text{--}n_{\text{H2}}$ degeneracy is a legitimate result. 5. **Systematics chapter (throughout).** Quantify how much of the measured ratio gradient could be mimicked by differential magnification, by residual uv -coverage mismatch, and by continuum-subtraction choices, using the source’s own multiply-imaged structure and by re-running the pipeline under alternative imaging weightings.

Deliverables. (i) A matched-beam, matched- uv multi-J cube set and ratio-map atlas for one lensed starburst; (ii) aperture-resolved non-LTE excitation constraints compared against the previously constrained global values, answering whether the global ratio is a luminosity-weighted average over strong internal gradients; (iii) a documented, scripted pipeline that the parent program can extend to the remaining 7 sources and eventually to pixel-by-pixel fitting.

Data and provenance notes. The idea uses only ADS/JAO.ALMA#2024.1.01396.S data, standard ALMA calibration/imaging tooling, and non-LTE escape-probability modelling — all described in the program itself; no new observations, instrumentation, or external catalogs are invoked. The data description names no alternative public dataset for these targets, so none is assumed. The description also does not state the redshift or observed sky frequency of any individual transition, only that the sample lies at $z = 1\text{--}3$ and that transitions are drawn from $J = 2\text{--}15$; consequently, the exact observed-frame frequency of each targeted line must be derived per source from its delivered spectral setup and verified against Splatalogue (using the source’s own redshift) before imaging, and no rest- vs. observed-frame value is assumed here. Likewise, the “four orders of magnitude in relative high-J vs. mid-J line ratios” is a sample-wide range quoted in the proposal, not a measurement for the selected target. If the delivered data for the chosen source are not yet accessible at the thesis start, the fallback is to develop and validate the identical matched-imaging + ratio + non-LTE pipeline on simulated cubes built from the program’s own stated observing parameters (0.1–0.2” beam, >10–20 beams across the source), so that the analysis runs immediately on delivery — not to substitute a different dataset.